Eason (BingXian) Xie

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Education

Northeastern University, Khoury College of Computer Science — Boston, MA

Bachelor of Science in Data Science and Mathematics — September 2024 – Expected May 2028

Data Science Concentration: Computer Science — **GPA: 3.67/4.0 | Dean's List**

Relevant Courses/Clubs: Object-Oriented Design, Database Design, Algorithms, Discrete Structures, Data Structures

Skills/Interests

Languages: Java, Python (Numpy, Pandas), JavaScript, Typescript, R, MySQL, VBA, DrRacket,

Technologies / Frameworks: Spring Boot, React, Node.js, Swing, NumPy, Pandas, D3.js, JUnit, MATLAB

Development Practices: Test-Driven Development, MVC Architecture, REST APIs, Pair Programming

Specializations/Data Science: Statistical Analysis, Data Visualization, Data Cleaning & Processing

Interests: Full-Stack Development, Open-Source Contribution, User Experience Design

Work Experience

City of Quincy, Inspectional Services | *Excel VBA, OpenGov, Data Analysis, Power BI* — Quincy, MA

Summer Intern — July 2022-August 2023

- Optimized municipal permit processing workflow by extracting and analyzing 10,000+ permit records from the OpenGov system, implementing data cleaning protocols and compliance pattern analysis that reduced processing time by 40% and improved departmental efficiency
- Enhanced citizen services and regulatory compliance by processing 100+ permit applications through comprehensive blueprint review, building code verification, and project progress tracking while serving as Chinese interpreter for 30+ non-English speaking applicants, improving accessibility and approval timelines
- Automated administrative operations and enhanced revenue tracking visibility by developing 10+ VBA macros for comprehensive report generation, enabling management to monitor fee revenue streams, track inspector productivity metrics, and assess departmental progress while eliminating processing bottlenecks and reducing manual errors across multiple permit categories

Project Experience

Construction Permit Management System | *SpringBoot, React, JSX* — *July 2025-March 2026*

Full-Stack Web Application (In Development)

- Addressed municipal permit processing inefficiencies identified during the City of Quincy internship, requiring a comprehensive digital solution to automate workflow and improve citizen accessibility
- Architecting a comprehensive permit management system using Java 21 Spring Boot backend and React JavaScript frontend with scalable multi-user architecture supporting role-based access control
- Building an industry-standard application workflow with a normalized database schema, comprehensive backend services, and a responsive frontend interface that addresses real-world municipal permit processing challenges

Multi-Calendar System: Supporting multiple calendar types | *Java* — *May 2025-Jun 2025*

- Engineered comprehensive calendar management solution for complex scheduling environments supporting multiple time zones, recurring events, and collaborative scheduling needs through advanced event management algorithms and data persistence optimization
- Developed a multi-calendar application using Java 11 with MVC architecture and implemented the Model layer with event management and data persistence using design patterns
- Delivered fully functional GUI application with complete CRUD operations using Swing framework, featuring intuitive navigation, advanced search functionality, and collaborative pair programming methodology to ensure code quality and user experience excellence

Carnegie Mellon University, Computer Science Department | *Machine Learning* — *Aug 2023-Oct 2023*

- Supported Music AI research by organizing and preprocessing large-scale audio datasets for machine learning model training, contributing to synthetic voice and music generation technology development through systematic data labeling, voice pattern recognition, and quality assurance protocols
- Advanced AI voice system development through comprehensive audio analysis and feature extraction including lyrics extraction, audio segmentation, phoneme timing annotation, and cross-genre feature study, enabling improved voice cloning and music generation algorithms with enhanced accuracy
- Enhanced training dataset quality by contributing 500+ processed soundtrack samples with systematic data preparation and audio analysis, directly supporting AI music generation and voice synthesis research